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QUESTION BANK

Semester – III Examinatation – 2026

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MAJ-4

CYBER SECURITY

PYTHON PROGRAMMING

UNIT I – Python File & Exception Handling

2 Marks

1. What is an Error in Python?
2. What is Exception Handling in Python?
3. What is the use of try block?
4. What is the use of except block?
5. What is Multiple Exception Handling?
6. What is a User-Defined Exception?
7. Define File Handling Mode.
8. What are the different File Handling Modes in Python?
9. What is the difference between Reading and Writing Files?
10. What is File Appending?
11. What is a File Exception?
12. What is the use of with statement in Python?

5/7 Marks

1. Explain Exception Handling in Python with suitable examples.
2. Explain different types of Errors in Python.
3. Explain the use of try, except, else and finally blocks with examples.
4. Explain Multiple Exception Handling in Python.
5. Explain how to create and raise User-Defined Exceptions.
6. Explain different File Handling Modes in Python with examples.
7. Explain how to read, write and append data to files in Python.
8. Explain File Exceptions and the use of with statement with examples.

UNIT II – Python Data Structures

2 Marks

1. What is List Comprehension?
2. What is Nested List Comprehension?
3. What is Dictionary Comprehension?
4. What is a Function in Python?
5. What are Default Parameters?
6. Define Variable Arguments.
7. What are Specialized Sorts in Python?
8. What is a Generator?
9. What is the use of yield statement?
10. What is a Lambda Expression?
11. What is the difference between *args and **kwargs?
12. What is a List in Python?

5/7 Marks

1. Describe Nested List Comprehension and Dictionary Comprehension.
2. Explain Lists in Python with suitable examples.
3. Describe Generators in Python. Explain yield with an example.
4. Write a detailed note on Lambda Expressions and their uses.

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5. Explain List Comprehension with examples.
6. Explain Nested List Comprehension with examples.
7. Explain Dictionary Comprehension with examples.
8. Explain Functions, Default Parameters and Variable Arguments with examples.
9. Explain Specialized Sorting Techniques in Python with examples.
10. Differentiate between *args and **kwargs with examples.

UNIT III – Collections

2 Marks

1. What is namedtuple()?
2. What is deque?
3. What is ChainMap?
4. What is Counter?
5. What is OrderedDict?
6. What is defaultdict?
7. What is UserDict?
8. What is UserList?
9. What is UserString?
10. What is the difference between List and Tuple?
11. What is the difference between Counter and defaultdict?

5/7 Marks

1. Explain namedtuple() with an example.
2. Explain deque and its operations with examples.
3. Explain ChainMap with an example.
4. Explain Counter with an example.
5. Explain OrderedDict and defaultdict with examples.
6. Explain UserDict, UserList and UserString.
7. Discuss the different types of Collections in Python with examples.
8. Differentiate between List, Tuple and Dictionary.

Advanced Features – 2 Marks

1. What is JSON?
2. What is JSON File Handling?
3. What is Excel File Handling in Python?
4. What is JSON Parsing?
5. What is the use of json module?

Advanced Features – 5/7 Marks

1. Explain JSON File Parsing and Handling in Python with examples.
2. Explain how to read and write JSON files in Python.
3. Explain Excel File Handling in Python with examples.
4. Discuss the importance of JSON and Excel files in data handling.

UNIT IV – Offensive Security with Python

2 Mraks

1. What is Offensive Security?
2. What is Cyber Attack?
3. What is Vulnerability?
4. What is Threat Modeling?
5. What is Penetration Testing?
6. What is Digital Forensics?
7. What is Security Operations Centre (SOC)?
8. What is Security Monitoring?
9. What is Logging?
10. What is Incident Response?

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11. What is an Incident Response Lifecycle?
12. What is a Security Protocol?

5/7 Marks

1. What are the different types of strategies for detecting a Cyber Attack?
2. How to design strategies for prevention and responding to Cyber Attacks? Explain briefly.
3. Describe the structure and daily operations of a modern Security Operations Centre (SOC).
4. Explain the major tools and strategies used for preventing, detecting and responding to Cyber Attacks.
5. Explain the process of planning and executing Penetration Tests.
6. Explain Threat Modeling and Vulnerability Analysis.
7. Explain Digital Forensics and its importance.
8. Explain Security Monitoring and Logging.
9. Describe the Incident Response Lifecycle.
10. Explain the strategies for implementing Security Protocols.

UNIT V – Basic Cryptography

2 Marks

1. What is Cryptography?
2. What is Symmetric Encryption?
3. What is Asymmetric Encryption?
4. What is a Hash Function?
5. What is SSL/TLS?
6. What is SSH?
7. What is IPsec?
8. What is Digital Signature?
9. What is Authentication Protocol?
10. What is Cryptanalysis?
11. What is the difference between Encryption and Decryption?
12. What is the difference between Symmetric and Asymmetric Encryption?

5/7 Marks

1. Write a short note on SSL/TLS, SSH and IPsec.
2. Explain the history and evolution of Cryptography.
3. Explain Symmetric Encryption with examples.
4. Explain Asymmetric Encryption with examples.
5. Differentiate between Symmetric and Asymmetric Encryption.
6. Explain Hash Functions and their applications.
7. Explain Classical and Modern Cryptanalysis Techniques.
8. Discuss different types of Attacks on Cryptographic Systems.
9. Explain Digital Signature and its working process.
10. Explain Authentication Protocols and their importance in Cyber Security.

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